

A Sign Classification of Total-Word Additive Level Sorts

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Abstract

Fix a finite nonempty alphabet, an integer weight for each letter, and either left-to-right or right-to-left scanning to break ties. The additive level sort orders the letters of a word by the sums of the weights strictly preceding their input positions. We prove that this function on all finite words is polyregular if and only if the weights are all nonnegative or all nonpositive. In every such case we give an arity-one first-order presentation. If both signs occur, a two-block input family makes the inverse image of a first-letter test nonregular, independently of the tie direction. Zero weights and arbitrary integer magnitudes are included. The classification requires agreement on all words; it does not settle the distinct problem of agreement only on Dyck words.

1 Statement and context

Let Σ be a fixed finite nonempty alphabet, let $\nu : \Sigma \rightarrow \mathbb{Z}$, and fix $d \in \{+1, -1\}$. For $w = w_1 \cdots w_n \in \Sigma^*$ define its starting levels by

$$\ell_w(i) = \sum_{j < i} \nu(w_j), \quad 1 \leq i \leq n. \quad (1)$$

The function $\Phi_{\nu,d} : \Sigma^* \rightarrow \Sigma^*$ lists every input position once, retaining its letter, in increasing lexicographic order of

$$(\ell_w(i), di). \quad (2)$$

We set $\Phi_{\nu,d}(\varepsilon) = \varepsilon$. Thus $d = +1$ scans equal levels from left to right, and $d = -1$ scans them from right to left. All comparisons below concern this total function, with agreement required on every input.

An arity-one first-order presentation uses the ordered input positions, unary letter predicates P_a , and fixed first-order formulas to select and label output positions and to order them. Our presentations select each input position once and keep its label. They are special cases of the tuple-MSO presentations of polyregular functions in [1, Definition 3.10]. Here MSO denotes monadic second-order logic on finite words. We use the standard fact that polyregular functions preserve regular languages under inverse image [2, Theorem 1.7]; the particular first-letter test needed below is also explained directly in Section 3.

Theorem 1. *For every fixed Σ , ν , and d as above, the following are equivalent:*

1. $\Phi_{\nu,d} \in \text{PolyReg}$ on all of Σ^* ;
2. $\Phi_{\nu,d}$ admits an arity-one first-order presentation;
3. either $\nu(a) \geq 0$ for every $a \in \Sigma$, or $\nu(a) \leq 0$ for every $a \in \Sigma$.

Baek, Hwang, La, and Yang define additive level sorts with these starting levels and scan ties in [1, Definition 5.2], using a binary alphabet. Their Section 11, problem (4), asks for a classification by

weights and tie order and points out the identity example with two weights equal to 1. Theorem 1 gives a sign criterion for the total-word interpretation, including arbitrary finite alphabets, together with explicit first-order formulas in every polyregular case.

The domain distinction matters. Realizing a map on a specified subset leaves the behavior outside that subset unconstrained [1, Definition 3.1]. The height-sweep lower bound in [1, Theorem 6.5] already rules out realization on Dyck words for its particular parameters. Our mixed-sign witnesses generally lie outside the Dyck domain. They give a uniform classification on Σ^* , without extending that stronger domain-specific lower bound to arbitrary weights. We do not consider input-dependent tie rules, state-dependent weights, or additional sorting layers. No equivalent sign classification was identified in the inspected passages of [1, 2]; this bounded comparison does not establish comprehensive novelty or priority.

2 First-order construction for one-sign weights

Let $Z(z)$ be the disjunction of $P_a(z)$ over letters with $\nu(a) = 0$; if there are no such letters, take $Z(z)$ to be $z \neq z$. With $\leq$ abbreviating $<$ or $=$, put

$$\begin{aligned} B(x, y, z) &:= (x \leq z \wedge z < y) \vee (y \leq z \wedge z < x), \\ E(x, y) &:= \forall z (B(x, y, z) \Rightarrow Z(z)). \end{aligned} \quad (3)$$

The interval includes its earlier endpoint and excludes its later endpoint. For $x < y$, equation (1) gives

$$\ell_w(y) - \ell_w(x) = \sum_{x \leq j < y} \nu(w_j). \quad (4)$$

If all weights have one weak sign, the sum vanishes exactly when every summand vanishes. Hence $E(x, y)$ is equivalent to $\ell_w(x) = \ell_w(y)$ for all positions x, y , including $x = y$.

In the nonnegative case define $L(x, y) := (x < y) \wedge \neg E(x, y)$; in the nonpositive case define $L(x, y) := (y < x) \wedge \neg E(x, y)$. In either case $L(x, y)$ expresses $\ell_w(x) < \ell_w(y)$. If all weights are zero, either definition gives the empty relation. Now put

$$\chi_d(x, y) := L(x, y) \vee (E(x, y) \wedge S_d(x, y)), \quad S_{+1}(x, y) := (x < y), \quad S_{-1}(x, y) := (y < x). \quad (5)$$

This is precisely strict comparison of the keys (2). Distinct positions have distinct keys, so χ_d is irreflexive, transitive, and total on distinct positions.

Use a single arity-one copy, the true domain sentence, selection formula $x = x$, label formulas $P_a(x)$, and output order (5). These are fixed first-order formulas once the parameters are fixed. They select each position once with its own letter in the required order. On empty input they select no positions. This proves (3) $\Rightarrow$ (2) in Theorem 1; the inclusion of these presentations in the polyregular class proves (2) $\Rightarrow$ (1).

3 A regular-preimage obstruction for mixed signs

Suppose that $\nu(a) = p > 0$ and $\nu(b) = -q < 0$ for two letters a, b , where p, q are positive integers. Restrict inputs to the regular language a^+b^+ . For $m, n \geq 1$, the a -levels of a^mb^n are $0, p, \dots, p(m-1)$, and its b -levels are $pm, pm - q, \dots, pm - q(n-1)$. The least a -level is 0 at the first position; the least b -level is $pm - q(n-1)$ at the final position. Both minima are unique within their letter blocks. Thus

$$\Phi_{\nu,d}(a^mb^n) \in b\Sigma^* \iff \begin{cases} q(n-1) > pm, & d = +1, \\ q(n-1) \geq pm, & d = -1. \end{cases} \quad (6)$$

At equality the first a and last b tie at level zero, explaining the two boundary conventions and the offset $n - 1$.

Let

$$K_d = \Phi_{\nu,d}^{-1}(b\Sigma^*) \cap a^+b^+. \quad (7)$$

For each integer $t \geq 1$ choose a prefix $x_t = a^{2qt}$ and a suffix $z_t = b^{2pt+2}$. On $x_s z_t$, the least b -level is

$$p(2qs) - q(2pt + 1) = q(2p(s - t) - 1). \quad (8)$$

For $s = t$ this equals $-q < 0$, whereas for $s > t$ it is at least $q > 0$. Consequently $x_t z_t \in K_d$ and $x_s z_t \notin K_d$ for every $s > t$, for either direction. Any two distinct prefixes are distinguished by the suffix indexed by the smaller index. The Myhill–Nerode theorem therefore implies that K_d is nonregular.

If $\Phi_{\nu,d}$ were polyregular, its inverse image of the regular language $b\Sigma^*$ would be regular [2, Theorem 1.7]. Intersection with a^+b^+ would make (7) regular, contradicting the preceding argument. This proves (1) $\Rightarrow$ (3) and completes Theorem 1.

For completeness, the required first-letter inverse-image fact can also be read directly from a tuple-MSO presentation. Let C be its finite set of copies, of fixed arities k_c , with domain dom , selection formulas sel_c , label formulas $\text{lab}_{c,b}$, and strict output-order formulas $\chi_{e,c}$. The sentence

$$\text{dom} \wedge \bigvee_{c \in C} \exists \bar{x} \left[\text{sel}_c(\bar{x}) \wedge \text{lab}_{c,b}(\bar{x}) \wedge \bigwedge_{e \in C} \forall \bar{y} (\text{sel}_e(\bar{y}) \Rightarrow \neg \chi_{e,c}(\bar{y}, \bar{x})) \right] \quad (9)$$

uses tuples of lengths k_c and k_e respectively. The conjunction and disjunction over copies are finite syntactic expansions. The sentence says that a selected output atom labeled b is the least output atom. It is false on empty output and correct on every finite nonempty strict total output order. It is an MSO sentence on the input word and hence defines a regular language by the MSO characterization of regular word languages. Only this first-letter property is needed for the obstruction.

4 Boundary cases and verification

The order formulas have a simple block interpretation. Partition a nonempty word into blocks consisting of zero-weight letters followed by one nonzero-weight letter, with an optional final nonempty block containing only zero-weight letters. An all-zero word forms a single block. These are the maximal intervals of equal starting level: the nonzero letter belongs at the end of its block because its step changes the level of the next position. For nonnegative weights, left-to-right ties give the identity, while right-to-left ties reverse each block in place. For nonpositive weights, left-to-right ties reverse the list of blocks while preserving each block internally, and right-to-left ties give reversal of the whole word. These descriptions follow from equations (3)–(5) and include trailing zeros. In particular, strictly positive weights give identity for both directions, strictly negative weights give reversal for both, and all-zero weights give identity or reversal according to the tie direction.

Corollary 2. *On a binary alphabet with weights in $\{-1, 0, 1\}$ and either scan direction, exactly four of the eighteen parameter choices are nonpolyregular: the two ordered mixed-sign weight pairs, each with either direction. The other fourteen admit arity-one first-order presentations.*

Proof. There are $3^2 \cdot 2 = 18$ choices. Theorem 1 excludes exactly the weight pairs $(1, -1)$ and $(-1, 1)$. $\square$

The proof is symbolic and has no bound on input length or on the fixed integer magnitudes. Supplementary exact Python checks compared the order formulas with direct level sorting and tested the threshold and prefix witness calculations. A separately implemented verifier checked 29,812 one-sign word/parameter/direction cases and 624,428 ordered position comparisons: all weak-sign weight vectors in $\{-4, -1, 0, 2, 5\}^2$ for lengths 0–7, and in $\{-3, 0, 7\}^4$ for lengths 0–4, with both directions and all ordered position pairs, including diagonals. It also checked 23,328 expanded mixed-sign cases with $1 \leq p, q \leq 9$, $1 \leq m, n \leq 12$, and both directions. These finite checks supplement the proof; the universal conclusions follow from the displayed identities and distinguishability argument. Zeng’s work on reversible cellular automata [3] provides broad adjacent context for symbolic analysis of word-encoded discrete systems accompanied by exact finite checks; it is not a premise of the level-sort classification.

Computational and AI assistance. OpenAI Codex was used in developing and checking the argument, implementing the exact computations, and preparing this manuscript. A separate model context reconstructed the proof and implemented the additional checks described above. This assistance is not external peer review, and no proof-assistant formalization of this theorem is supplied.

References

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